

REAL ANALYSIS

Tutorial 1

'How natural are the Natural Numbers?'

Peano Axioms and the Well Ordering Principle

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What is the Natural Numbers?

“God made the integers, all else is the work of man.” — Leopold Kronecker

OUR INTUITIVE NOTIONS OF $\mathbb{N}$:

1. Endless Quantity

$0, 1, 2, 3, \dots$
 $\dots, 12131131, 12131132, \dots$

- There are an **endless number** of elements.
- No “largest” natural number exists.

2. Arithmetic Structure

$5 + 7 = 12$ $45 + 34 = 79$
 $20 \times 2 = 40$ $12 \times 3 = 36$

- Operations of **addition** (+) and **multiplication** ($\times$).
- Governed by algebraic properties.

3. Natural Ordering

$4 < 7$
 $2000 < 300000$

- A definitive **ordering** relation ($<$).
- Some numbers are strictly “greater” than others.

What is the Natural Numbers?

FORMALIZATIONS:

Set-Theoretic Notions: $0 = \emptyset, \quad S(a) = a \cup \{a\}$

(Constructed using the Axiom of Pairwise Union, Axiom of Infinity, etc.)

Peano Axioms

P1. $0 \in \mathbb{N}$

P2. $n \in \mathbb{N} \implies S(n) \in \mathbb{N}$

P3. $S(m) = S(n) \implies m = n$

P4. $S(n) \neq 0, \forall n \in \mathbb{N}$

P5. $(0 \in K \wedge S(K) \subseteq K) \implies K = \mathbb{N}$

Well Ordering Principle



Peano's Axioms

Formulated by **Giuseppe Peano** to define $\mathbb{N}$ using primitive elements: a base element 0 and a unary successor map $S(n) = n++$ (intuitive next element).

P1 & P2: Genesis & Succession

P1. $0 \in \mathbb{N}$

In this formalization, 0 is explicitly designated as a natural number.

P2. If $n \in \mathbb{N}$, then $n++ \in \mathbb{N}$

Closure under succession; allows an indefinite chain of numbers starting from 0.

P3: A Strict Starting Point

P3. $n++ \neq 0$ for all $n \in \mathbb{N}$

0 is never a successor. This eliminates finite loops (e.g., modular arithmetic).

Key Distinction:

“0 is not a successor” $\neq$ “Every non-zero is a successor”.

Peano's Axioms:

P4: Injectivity of Succession

Statement:

If $n++ = m++$, then $n = m$.

Distinct numbers have distinct successors; chains never merge together.

Contrapositive Formulation:

If $n \neq m$, then $n++ \neq m++$.

P5: Principle of Mathematical Induction

[CRITICAL FOUNDATION]

Let $P(n)$ be a property of numbers $n \in \mathbb{N}$. If:

- **Base Case:** $P(0)$ is true
- **Inductive Step:** $P(n) \implies P(n++)$

$\implies P(n)$ holds for all $n \in \mathbb{N}$.

The Minimality Constraint:

P1–P4 prevent loops and branch-merging, but still permit 'junk' copies (e.g., extra disjoint copies of $\mathbb{Z}$).

P5 forces minimality: $\mathbb{N}$ contains *only* objects generated starting from 0 via iterated succession.

Linear Order Relation ($>$)

The relation $>$ is a **binary relation**: it acts on pairs of elements in a set M .

The Three Order Properties

A relation $<$ on M is a **strict linear order** if for all $x, y, z \in M$:

- L1. Irreflexivity:** $x < x$ is false.
- L2. Transitivity:** If $x < y$ and $y < z$, then $x < z$.
- L3. Trichotomy:** Exactly one of $x < y$, $x = y$, or $y < x$ holds.

If such an order is defined on all elements of M , $(M, <)$ is a **linearly ordered set**.

Standard Notation:

- $x \leq y$ denotes " $x < y$ or $x = y$ "
- $y > x$ means $x < y$

Intuition

Provides a systematic comparison between elements:

Definite Order $\implies$ **'Linear'** $\implies$ **Line**

Least Element

Definition 2: Least Element

Let $(M, <)$ be a linearly ordered set and let $A \subseteq M$. An element $m \in M$ is a **least element** of A if:

$$m \in A \quad \text{and} \quad m \leq a \quad \text{for every } a \in A.$$

Uniqueness

A subset A of a linearly ordered set has **at most one** least element.

Proof:

- Suppose both m and m' are least elements of A .
- Since $m' \in A$ and m is a least element, we have $m \leq m'$.
- Symmetrically, since $m \in A$ and m' is a least element, we have $m' \leq m$.
- By antisymmetry (derived from irreflexivity and trichotomy), $m \leq m'$ and $m' \leq m$ implies $m = m'$. □

The Well-Ordering Principle

Well-Ordering Principle

A linearly ordered set $(M, <)$ is **well-ordered** if **every non-empty subset has a least element**:

$$\forall A \subseteq M, A \neq \emptyset \implies \exists m \in A \text{ such that } \forall a \in A (m \leq a).$$

Instances of the Principle in $\mathbb{N}$

- **The entire set:** The set $\mathbb{N}$ itself is non-empty; its least element is 0.
- **Non-zero elements:** For $A = \{n \in \mathbb{N} : n \neq 0\}$, the least element is 0++.
- **Least Counterexample:** Let $A = \{n \in \mathbb{N} : P(n) \text{ is false}\}$. Consider a P that is not true for all n , then elements for which it fails is in A , $A \neq \emptyset$, and the principle yields a **least counterexample** $m \in \mathbb{N}$ (i.e., $P(m)$ fails, but $P(k)$ holds for all $k < m$).

When Well-Ordering Fails

Well-ordering is **not** a generic 'obvious' property of linearly ordered sets; there are quite a lot of examples for which it fails

Non-Example 1: Descending Chain

Let $D = \{d_0, d_1, d_2, \dots\}$ be ordered by:

$$\dots < d_3 < d_2 < d_1 < d_0$$

- $d_j < d_i \iff i < j$.
- Can be shown to be linearly ordered
- D is non-empty, but has **no least element**: for any candidate d_i , $d_{i+1} < d_i$.
- Hence, $(D, <)$ is **not** well ordered

Non-Example 2: Bounded Below

Let $M = \{b\} \cup D$ with D as before, and set $b < d_i$ for all i :

- Consider the subset $D \subseteq M$.
- D is bounded below in M by b .
- However, D still has no least element because $b \notin D$.
- **A lower bound in M does not guarantee a least element in A .**

Some Basics

Induction (P5) $\implies$ Recursive Defns $\implies$ Addition (+) $\implies$ Order (<)

*Order properties are not assumed—they are **theorems** proved via addition and induction.*

Fundamental Properties of Order on $\mathbb{N}$ (via Induction)

Using Mathematical Induction, we establish three foundational order theorems:

F1. Minimal Base Element: $0 \leq n$ for every $n \in \mathbb{N}$

F2. Successor Compatibility: $n < a \iff n++ \leq a$ for all $a, n \in \mathbb{N}$

F3. Trichotomy: For all $a, b \in \mathbb{N}$, exactly one holds:

$$a < b, \quad a = b, \quad b < a$$

Proving Well-Ordering for $\mathbb{N}$

We showed earlier that well-ordering holds for particular subsets of $\mathbb{N}$. But the statement says it should hold for **every** subset. Let's prove that the Well-Ordering Principle holds for $\mathbb{N}$! For this, we **only** use Peano's axioms and its consequents. **Nothing else is assumed.**

Theorem: Well-Ordering of $\mathbb{N}$

Assume Peano axioms **P1–P5** and order facts **(F1)–(F3)**. Every non-empty subset $A \subseteq \mathbb{N}$ has a least element.

Proof Strategy: Contradiction via Induction

Let $A \subseteq \mathbb{N}$ be non-empty. Suppose for contradiction that A **has no least element**.

Define the property:

$$P(n) : \text{no element of } A \text{ is } \leq n \quad (\text{i.e., } \forall a \in A, a > n)$$

Goal: Prove that $P(n)$ holds for **every** $n \in \mathbb{N}$ using induction (**P5**).

Proving Well-Ordering for $\mathbb{N}$

Base Case: Prove $P(0)$

Suppose some $a \in A$ satisfies $a \leq 0$.

- By **(F1)**, $0 \leq a$. Hence $a = 0 \implies 0 \in A$.
- Since $0 \leq x$ for all $x \in \mathbb{N}$, 0 is a lower bound for A .
- Being inside A , 0 is a **least element** of A —contradicting our assumption!

$\implies$ No element of A is ≤ 0 . $P(0)$ **holds**.

Inductive Step: $P(n) \implies P(n++)$

Assume $P(n)$ holds: no element in A is $\leq n$.

Suppose for contradiction that some $a \in A$ has $a \leq n++$:

- (i) $a \not\leq n \implies n < a$ by Trichotomy **(F3)**.
- (ii) By **(F2)**, $n < a \implies n++ \leq a$.
- (iii) Since $n++ \leq a \leq n++$, we get $a = n++ \in A$.

For any $x \in A$, $x \not\leq n \implies n++ \leq x$. Thus $n++$ is a lower bound in $A \implies$ **least element** ($\perp$). $\implies P(n++)$ **holds**.

Proving Well-Ordering for $\mathbb{N}$

Conclusion via Induction (Axiom P5)

Since both the base case and inductive step hold, Axiom **P5** guarantees:

$P(n)$ holds for every $n \in \mathbb{N}$

The Contradiction

- Since $A \neq \emptyset$, choose an arbitrary element $a_0 \in A$.
- Applying $P(n)$ at $n = a_0$: **“No element of A is $\leq a_0$ ”**.
- But $a_0 \in A$ and $a_0 \leq a_0$ (reflexivity), directly contradicting $P(a_0)$!

The supposition fails: **every non-empty subset of $\mathbb{N}$ has a least element.**

Thus, $(\mathbb{N}, <)$ is a **well-ordered set**.



Swapping Induction and Well Ordering!

What We Just Proved: Peano Axioms (P1–P4 + **P5 [Induction]**) $\implies$ **Well-Ordering Principle**.

The Thought Experiment: Tweaking Peano

The axioms we choose are not set in stone—they are foundational design decisions.

- What if we discard **P5 (Induction)** and replace it directly with the **Well-Ordering Principle**?
- **New Axiom Set:** P1–P4 + **Well-Ordering Principle**.

The Test of Equivalence

- **The Core Question:** How can we be certain this newly axiomatized set is genuinely our familiar $\mathbb{N}$?
- Induction is intuitive and foundational to arithmetic on natural numbers.
- **The Verification:** To guarantee we describe the exact same structure, we must now prove the converse:

Peano (w/ Well-Ordering) $\implies$ Principle of Mathematical Induction

Proving Induction from Well-Ordering

Hypothesis & Set of Counterexamples

Let P be a property of $\mathbb{N}$ such that: (1) $P(0)$ is true, and (2) $P(n) \implies P(n++)$ for all $n \in \mathbb{N}$.

Define the set of counterexamples: $A = \{n \in \mathbb{N} : P(n) \text{ is false}\}$

We aim to show $A = \emptyset$. Suppose for contradiction that $A \neq \emptyset$.

Constructing the Predecessor

- **Step 1 (Minimal Counterexample):** By the Well-Ordering Principle, non-empty A has a least element $m \in A$, so $m \leq a$ for all $a \in A$.
- **Step 2 ($m \neq 0$):** Since hypothesis (1) gives $P(0)$ is true, $0 \notin A$. Because $m \in A$, we have $m \neq 0$.
- **Step 3 (Immediate Predecessor):** Since $m \neq 0$, m must be a successor. Hence, there exists some $p \in \mathbb{N}$ such that $p++ = m$.
- **Step 4 (Predecessor is Smaller):** Since $p < p++$ and $p++ = m$, we obtain $p < m$.

Proving Induction from Well-Ordering

Step 5: The Predecessor is NOT a Counterexample

Could $p \in A$? If so, minimality of m requires $m \leq p$:

- If $m = p \implies p < p$, which violates irreflexivity.
- If $m < p \implies$ combined with $p < m$, gives $m < m$ by transitivity (impossible).

$\implies p \notin A$. Therefore, $P(p)$ **must be true!**

Step 6: Deriving the Contradiction

- By hypothesis (2), $P(p) \implies P(p++)$ is true.
- Since $p++ = m$, this means $P(m)$ **is true**, so $m \notin A$.
- This directly contradicts Step 1 ($m \in A$).

The supposition $A \neq \emptyset$ fails $\implies A = \emptyset$. Hence, $P(n)$ **holds for all** $n \in \mathbb{N}$.

Are Induction and Well-Ordering Interchangeable?

Have we just shown that **Induction** and the **Well-Ordering Principle** are fully interchangeable?

Not quite! ... because the previous proof contains a fatal flaw.

Where Did We Go Wrong?

Look back closely at the steps in that derivation.

- Every deduction seemed algebraically valid on the surface.
- Yet, at least one step secretly invoked a fact we were **not yet entitled to use**.

Hint: Beware of Circularity

Think about circular reasoning (*Petitio Principii*).

Did we rely on a property of succession or order whose *own* proof secretly requires **Induction**?

The Problems

Problem 1

To even state the Well-Ordering Principle on $\mathbb{N}$, one needs the relation $\leq$. But $\leq$ is **not a primitive** in Peano arithmetic—its construction proceeds via addition and recursion, both of which require **Induction (P5)**! Without induction, well-ordering is not even a well-formed statement unless $<$ is taken as an additional primitive.

Problem 2

- In Step 4, we assumed $p < p++$.
- In Peano arithmetic, $<$ is defined via addition, addition via recursion, and recursion is justified strictly by **Induction**.
- Using $p < p++$ to prove induction is completely **circular**.

Problem 3

- In Step 3, we assumed every $m \neq 0$ has a predecessor p ($p++ = m$).
- **P3** states $n++ \neq 0$, which does *not* imply every non-zero is a successor.
- This fact is itself a theorem proved via **Induction**—another fatal circularity!

Fixing the System: An Augmented Axiom Set

The Diagnosis: The Well-Ordering Principle alone cannot derive Induction over $(0, ++)$. Without induction, the system permits models strictly **larger than** $\mathbb{N}$ with unconstrained stray elements.

The Remedy: Upgrade the Gaps to Axioms

To resolve the gaps, we introduce four explicit axioms:

- (A) Primitive Linear Order:** $<$ is a primitive relation satisfying irreflexivity (L1), transitivity (L2), and trichotomy (L3).
- (W) Well-Ordering:** Every non-empty subset of $\mathbb{N}$ has a least element under $<$.
- (S) Predecessors Exist:** For every $n \neq 0$, there exists $p \in \mathbb{N}$ such that $p++ = n$.
- (O) Successors Increase:** $n < n++$ for every $n \in \mathbb{N}$.

Student Exercise

Homework: Repair the flawed proof (Slides 15–16) using axioms **(A)**, **(W)**, **(S)**, and **(O)**.

Is Well-Ordering Actually Weaker?

Within the bare Peano signature $(0, ++)$: **Yes.** In general: **No!**

If we shift perspective and take **order as primary**, we can build a complete, alternate foundation.

Pure Order-Theoretic Axioms on M

- (O1) Linear Order:** $<$ is a strict linear order on a non-empty set M .
- (O2) Well-Ordering:** Every non-empty subset of M has a least element.
- (O3) No Maximum:** M has no greatest element.
- (O4) Order Completeness:** Every non-empty subset of M bounded above has a greatest element.

Recovering Arithmetic Notions

We can now *define* the Peano primitives directly:

- 0 := least element of M
(Well-defined by **O2** since $M \neq \emptyset$)
- $n++$:= least element of $\{x \in M : n < x\}$
(**O3** ensures the set is non-empty; **O2** yields a min)

An Alternate Foundation

From **(O1)–(O4)**, one can rigorously prove **all five Peano axioms**—including Induction **(P5)**!